EXERCISE 4

15 Second-Order Homogeneous Linear ODEs with Initial Conditions

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Characteristic Polynomial • 15 IVPs Solved

PHASE 1: STATEMENT & SETUP

Find the general solution of the following differential equations, and then the unique solution satisfying the specified initial conditions:

(i) $\ddot{x} - 3\dot{x} + 2x = 0,\ x(0) = 2, \dot{x}(0) = 6$

(ii) $y'' - 4y' + 4y = 0,\ y(0) = 0, y'(0) = 3$

(iii) $z'' - 4z' + 13z = 0,\ z(0) = 7, z'(0) = 42$

(iv) $\ddot{y} + \dot{y} - 6y = 0,\ y(0) = -1, y'(0) = 8$

(v) $\ddot{y} - 4\dot{y} = 0,\ y(0) = 13, \dot{y}(0) = 0$

(vi) $\ddot{\theta} + 4\theta = 0,\ \theta(0) = 0, \dot{\theta}(0) = 10$

(vii) $\ddot{y} + 2\dot{y} + 10y = 0,\ y(0) = 3, \dot{y}(0) = 0$

(viii) $2\ddot{z} + 7\dot{z} - 4z = 0,\ z(0) = 0, \dot{z}(0) = 9$

(ix) $\ddot{y} + 2\dot{y} + y = 0,\ y(0) = 0, \dot{y}(0) = -1$

(x) $\ddot{x} + 6\dot{x} + 10x = 0,\ x(0) = 3, \dot{x}(0) = 1$

(xi) $4\ddot{x} - 20\dot{x} + 21x = 0,\ x(0) = -4, \dot{x}(0) = -12$

(xii) $\ddot{y} + \dot{y} - 2y = 0,\ y(0) = 4, \dot{y}(0) = -4$

(xiii) $\ddot{y} - 4y = 0,\ y(0) = 10, \dot{y}(0) = 0$

(xiv) $y'' + 4y' + 4y = 0,\ y(0) = 27, y'(0) = -54$

(xv) $\ddot{y} + \omega^2 y = 0,\ y(0) = 0, \dot{y}(0) = 1$

PHASE 2: FUNDAMENTAL FORMULATION

For $a\ddot{u} + b\dot{u} + cu = 0$, the exponential ansatz $u(t) = e^{rt}$ yields the auxiliary equation:

$$ar^2 + br + c = 0, \qquad \Delta = b^2 - 4ac$$

- $\Delta > 0$: $u(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$ (Distinct Real).
- $\Delta = 0$: $u(t) = (c_1 + c_2 t)e^{rt}$ (Repeated Real).
- $\Delta < 0$: $u(t) = e^{\rho t}(c_1 \cos \omega t + c_2 \sin \omega t)$ (Complex Conjugate $r = \rho \pm i\omega$).

Constants c_1, c_2 are uniquely determined by Cauchy conditions $u(0) = u_0, \dot{u}(0) = v_0$.

METHOD OF THE CLASS NOTES [NOTES, P. 5]

The notes divide by the leading coefficient to reach $x'' + px' + qx = 0$ (items 8 and 11 need it), try $x = e^{kt}$, get $k^2 + pk + q = 0, k = \frac{-p \pm \sqrt{p^2 - 4q}}{2}$, and read the case: Case 01, $k_1 \neq k_2$ real, $C_1 e^{k_1 t} + C_2 e^{k_2 t}$; Case 02, double root $k = -p/2, C_1 e^{kt} + C_2 t e^{kt}$; Case 03, $\rho \pm \omega i$ with $\rho = -p/2, \omega = \sqrt{4q - p^2}/2, e^{\rho t}(A \cos \omega t + B \sin \omega t)$.

The initial data are then matched as in the example of the notes:

$$\begin{pmatrix} x_1(0) & x_2(0) \\ x_1'(0) & x_2'(0) \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} x(0) \\ x'(0) \end{pmatrix},$$

a system whose determinant is the Wronskian $W(0) \neq 0$, so every item has exactly one solution.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION (ALL 15 ITEMS)

(i) $\ddot{x} - 3\dot{x} + 2x = 0$ **with** $x(0) = 2,\ \dot{x}(0) = 6$

Auxiliary eq: $r^2 - 3r + 2 = 0 \implies (r-1)(r-2) = 0 \implies r_1 = 1, r_2 = 2.$

General solution: $x(t) = c_1 e^t + c_2 e^{2t} \implies \dot{x}(t) = c_1 e^t + 2c_2 e^{2t}.$

Initial conditions: $c_1 + c_2 = 2, c_1 + 2c_2 = 6 \implies c_2 = 4,\ c_1 = -2.$

Solution: $\mathbf{x(t) = -2e^t + 4e^{2t}}$

(ii) $y'' - 4y' + 4y = 0$ **with** $y(0) = 0,\ y'(0) = 3$

Auxiliary eq: $r^2 - 4r + 4 = 0 \implies (r-2)^2 = 0 \implies r = 2$ (mult. 2).

General solution: $y(t) = (c_1 + c_2 t)e^{2t} \implies y'(t) = (2c_1 + c_2 + 2c_2 t)e^{2t}.$

Initial conditions: $y(0) = c_1 = 0; y'(0) = 2(0) + c_2 = 3 \implies c_2 = 3.$

Solution: $\mathbf{y(t) = 3te^{2t}}$

(iii) $z'' - 4z' + 13z = 0$ **with** $z(0) = 7,\ z'(0) = 42$

Auxiliary eq: $r^2 - 4r + 13 = 0 \implies (r-2)^2 + 9 = 0 \implies r = 2 \pm 3i\ (\rho = 2, \omega = 3)$.

General solution: $z(t) = e^{2t}(c_1 \cos 3t + c_2 \sin 3t)$.

Derivative: $z'(t) = e^{2t}\left[(2c_1 + 3c_2)\cos 3t + (2c_2 - 3c_1)\sin 3t\right]$.

Initial conditions: $z(0) = c_1 = 7$; $z'(0) = 2(7) + 3c_2 = 42 \implies 3c_2 = 28 \implies c_2 = \frac{28}{3}$.

Solution: $\mathbf{z(t) = e^{2t}\left(7\cos 3t + \frac{28}{3}\sin 3t\right)}$

(iv) $\ddot{y} + \dot{y} - 6y = 0$ **with** $y(0) = -1,\ y'(0) = 8$

Auxiliary eq: $r^2 + r - 6 = 0 \implies (r+3)(r-2) = 0 \implies r_1 = 2, r_2 = -3$.

General solution: $y(t) = c_1 e^{2t} + c_2 e^{-3t} \implies \dot{y}(t) = 2c_1 e^{2t} - 3c_2 e^{-3t}$.

Initial conditions: $c_1 + c_2 = -1, 2c_1 - 3c_2 = 8 \implies 2c_1 - 3(-1 - c_1) = 5c_1 + 3 = 8 \implies c_1 = 1,\ c_2 = -2$.

Solution: $\mathbf{y(t) = e^{2t} - 2e^{-3t}}$

(v) $\ddot{y} - 4\dot{y} = 0$ **with** $y(0) = 13,\ \dot{y}(0) = 0$

Auxiliary eq: $r^2 - 4r = 0 \implies r(r-4) = 0 \implies r_1 = 0, r_2 = 4$.

General solution: $y(t) = c_1 + c_2 e^{4t} \implies \dot{y}(t) = 4c_2 e^{4t}$.

Initial conditions: $\dot{y}(0) = 4c_2 = 0 \implies c_2 = 0$; $y(0) = c_1 = 13$.

Solution: $\mathbf{y(t) = 13}$ (Steady constant equilibrium)

(vi) $\ddot{\theta} + 4\theta = 0$ **with** $\theta(0) = 0,\ \dot{\theta}(0) = 10$

Auxiliary eq: $r^2 + 4 = 0 \implies r = \pm 2i\ (\rho = 0, \omega = 2)$.

General solution: $\theta(t) = c_1 \cos 2t + c_2 \sin 2t \implies \dot{\theta}(t) = -2c_1 \sin 2t + 2c_2 \cos 2t$.

Initial conditions: $\theta(0) = c_1 = 0$; $\dot{\theta}(0) = 2c_2 = 10 \implies c_2 = 5$.

Solution: $\theta(\mathbf{t}) = \mathbf{5}\sin \mathbf{2t}$

(vii) $\ddot{y} + 2\dot{y} + 10y = 0$ **with** $y(0) = 3,\ \dot{y}(0) = 0$

Auxiliary eq: $r^2 + 2r + 10 = 0 \implies (r+1)^2 + 9 = 0 \implies r = -1 \pm 3i\ (\rho = -1, \omega = 3)$.

General solution: $y(t) = e^{-t}(c_1 \cos 3t + c_2 \sin 3t)$.

Derivative: $\dot{y}(t) = e^{-t}\left[(3c_2 - c_1)\cos 3t - (3c_1 + c_2)\sin 3t\right]$.

Initial conditions: $y(0) = c_1 = 3$; $\dot{y}(0) = 3c_2 - 3 = 0 \implies c_2 = 1$.

Solution: $\mathbf{y(t) = e^{-t}\left(3\cos 3t + \sin 3t\right)}$

(viii) $2\ddot{z} + 7\dot{z} - 4z = 0$ **with** $z(0) = 0,\ \dot{z}(0) = 9$

Auxiliary eq: $2r^2+7r-4=0 \implies (2r-1)(r+4)=0 \implies r_1=\frac{1}{2}, r_2=-4$.

General solution: $z(t)=c_1e^{t/2}+c_2e^{-4t} \implies \dot{z}(t)=\frac{1}{2}c_1e^{t/2}-4c_2e^{-4t}$.

Initial conditions: $z(0)=c_1+c_2=0 \implies c_2=-c_1; \dot{z}(0)=\frac{1}{2}c_1+4c_1=\frac{9}{2}c_1=9 \implies c_1=2, c_2=-2$.

Solution: $\mathbf{z(t)=2e^{t/2}-2e^{-4t}}$

(ix) $\ddot{y}+2\dot{y}+y=0$ **with** $y(0)=0,\ \dot{y}(0)=-1$

Auxiliary eq: $r^2+2r+1=0 \implies (r+1)^2=0 \implies r=-1$ (mult. 2).

General solution: $y(t)=(c_1+c_2t)e^{-t} \implies \dot{y}(t)=(c_2-c_1-c_2t)e^{-t}$.

Initial conditions: $y(0)=c_1=0; \dot{y}(0)=c_2-0=-1 \implies c_2=-1$.

Solution: $\mathbf{y(t)=-te^{-t}}$

(x) $\ddot{x}+6\dot{x}+10x=0$ **with** $x(0)=3,\ \dot{x}(0)=1$

Auxiliary eq: $r^2+6r+10=0 \implies (r+3)^2+1=0 \implies r=-3\pm i\ (\rho=-3, \omega=1)$.

General solution: $x(t)=e^{-3t}(c_1\cos t+c_2\sin t)$.

Derivative: $\dot{x}(t)=e^{-3t}\left[(c_2-3c_1)\cos t-(c_1+3c_2)\sin t\right]$.

Initial conditions: $x(0)=c_1=3; \dot{x}(0)=c_2-3(3)=1 \implies c_2=10$.

Solution: $\mathbf{x(t)=e^{-3t}\left(3\cos t+10\sin t\right)}$

(xi) $4\ddot{x}-20\dot{x}+21x=0$ **with** $x(0)=-4,\ \dot{x}(0)=-12$

Auxiliary eq: $4r^2-20r+21=0. \Delta=400-336=64=8^2 \implies r=\frac{20\pm8}{8} \implies r_1=\frac{7}{2}, r_2=\frac{3}{2}$.

General solution: $x(t)=c_1e^{3t/2}+c_2e^{7t/2} \implies \dot{x}(t)=\frac{3}{2}c_1e^{3t/2}+\frac{7}{2}c_2e^{7t/2}$.

Initial conditions: $c_1+c_2=-4, \frac{3}{2}c_1+\frac{7}{2}(-4-c_1)=-2c_1-14=-12 \implies -2c_1=2 \implies c_1=-1,\ c_2=-3$.

Solution: $\mathbf{x(t)=-e^{3t/2}-3e^{7t/2}}$

(xii) $\ddot{y}+\dot{y}-2y=0$ **with** $y(0)=4,\ \dot{y}(0)=-4$

Auxiliary eq: $r^2+r-2=0 \implies (r+2)(r-1)=0 \implies r_1=1, r_2=-2$.

General solution: $y(t)=c_1e^{t}+c_2e^{-2t} \implies \dot{y}(t)=c_1e^{t}-2c_2e^{-2t}$.

Initial conditions: $c_1+c_2=4, c_1-2c_2=-4 \implies 3c_2=8 \implies c_2=\frac{8}{3},\ c_1=\frac{4}{3}$.

Solution: $\mathbf{y(t)=\frac{4}{3}e^{t}+\frac{8}{3}e^{-2t}}$

(xiii) $\ddot{y}-4y=0$ **with** $y(0)=10,\ \dot{y}(0)=0$

Auxiliary eq: $r^2-4=0 \implies r=\pm2$.

General solution: $y(t)=c_1e^{2t}+c_2e^{-2t} \implies \dot{y}(t)=2c_1e^{2t}-2c_2e^{-2t}$.

Initial conditions: $c_1 + c_2 = 10, 2(c_1 - c_2) = 0 \implies c_1 = c_2 = 5$.

Solution: $\mathbf{y(t) = 5e^{2t} + 5e^{-2t} = 10\cosh(2t)}$

(xiv) $y'' + 4y' + 4y = 0$ **with** $y(0) = 27,\ y'(0) = -54$

Auxiliary eq: $r^2 + 4r + 4 = 0 \implies (r+2)^2 = 0 \implies r = -2$ (mult. 2).

General solution: $y(t) = (c_1 + c_2 t)e^{-2t} \implies y'(t) = (c_2 - 2c_1 - 2c_2 t)e^{-2t}$.

Initial conditions: $y(0) = c_1 = 27$; $y'(0) = c_2 - 2(27) = -54 \implies c_2 = 0$.

Solution: $\mathbf{y(t) = 27e^{-2t}}$

(xv) $\ddot{y} + \omega^2 y = 0$ **with** $y(0) = 0,\ \dot{y}(0) = 1$ **(**$\omega \neq 0$**)**

Auxiliary eq: $r^2 + \omega^2 = 0 \implies r = \pm i\omega$.

General solution: $y(t) = c_1\cos(\omega t) + c_2\sin(\omega t) \implies \dot{y}(t) = -c_1\omega\sin(\omega t) + c_2\omega\cos(\omega t)$.

Initial conditions: $y(0) = c_1 = 0$; $\dot{y}(0) = c_2\omega = 1 \implies c_2 = \frac{1}{\omega}$.

Solution: $\mathbf{y(t) = \frac{1}{\omega}\sin(\omega t)}$

PHASE 4: MASTER SUMMARY TABLE

Item	Differential Equation	Initial Conditions	Roots r	Unique IVP Solution
(i)	$\ddot{x} - 3\dot{x} + 2x = 0$	$x(0) = 2, \dot{x}(0) = 6$	$1, 2$	$\mathbf{-2e^{t} + 4e^{2t}}$
(ii)	$y'' - 4y' + 4y = 0$	$y(0) = 0, y'(0) = 3$	2 (mult. 2)	$\mathbf{3te^{2t}}$
(iii)	$z'' - 4z' + 13z = 0$	$z(0) = 7, z'(0) = 42$	$2 \pm 3i$	$\mathbf{e^{2t}(7\cos 3t + \frac{28}{3}\sin 3t)}$
(iv)	$\ddot{y} + \dot{y} - 6y = 0$	$y(0) = -1, y'(0) = 8$	$2, -3$	$\mathbf{e^{2t} - 2e^{-3t}}$
(v)	$\ddot{y} - 4\dot{y} = 0$	$y(0) = 13, \dot{y}(0) = 0$	$0, 4$	$\mathbf{13}$
(vi)	$\ddot{\theta} + 4\theta = 0$	$\theta(0) = 0, \dot{\theta}(0) = 10$	$\pm 2i$	$\mathbf{5\sin 2t}$
(vii)	$\ddot{y} + 2\dot{y} + 10y = 0$	$y(0) = 3, \dot{y}(0) = 0$	$-1 \pm 3i$	$\mathbf{e^{-t}(3\cos 3t + \sin 3t)}$
(viii)	$2\ddot{z} + 7\dot{z} - 4z = 0$	$z(0) = 0, \dot{z}(0) = 9$	$\frac{1}{2}, -4$	$\mathbf{2e^{t/2} - 2e^{-4t}}$
(ix)	$\ddot{y} + 2\dot{y} + y = 0$	$y(0) = 0, \dot{y}(0) = -1$	-1 (mult. 2)	$\mathbf{-te^{-t}}$
(x)	$\ddot{x} + 6\dot{x} + 10x = 0$	$x(0) = 3, \dot{x}(0) = 1$	$-3 \pm i$	$\mathbf{e^{-3t}(3\cos t + 10\sin t)}$
(xi)	$4\ddot{x} - 20\dot{x} + 21x = 0$	$x(0) = -4, \dot{x}(0) = -12$	$\frac{3}{2}, \frac{7}{2}$	$\mathbf{-e^{3t/2} - 3e^{7t/2}}$
(xii)	$\ddot{y} + \dot{y} - 2y = 0$	$y(0) = 4, \dot{y}(0) = -4$	$1, -2$	$\mathbf{\frac{4}{3}e^{t} + \frac{8}{3}e^{-2t}}$
(xiii)	$\ddot{y} - 4y = 0$	$y(0) = 10, \dot{y}(0) = 0$	± 2	$\mathbf{10\cosh 2t}$
(xiv)	$y'' + 4y' + 4y = 0$	$y(0) = 27, y'(0) = -54$	-2 (mult. 2)	$\mathbf{27e^{-2t}}$
(xv)	$\ddot{y} + \omega^2 y = 0$	$y(0) = 0, \dot{y}(0) = 1$	$\pm i\omega$	$\mathbf{\frac{1}{\omega}\sin(\omega t)}$